

DSA0109 – C++ Programming

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1. Factorial

Factorial-Fun. ▾

FACTORIAL-FUN.

Write a function to compute the factorial of a given integer.

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int n, fact=1;
6     cin>>n;
7     for (int i=1; i<= n; i++)
8         fact=fact*i;
9     cout<<fact;
10    return 0;
11 }
```

5

OUTPUT

120

2. Prime

Prime-Fun. ▾

PRIME-FUN.

Write a function to determine if a given integer is a prime number or not.

```
1 #include<iostream>
2 using namespace std;
3 int main() {
4     int n,i;
5     cin>>n;
6     for(i=2;i<n;i++)
7     {
8         if(n%i==0)
9         {
10             cout<<"Not prime";
11             return 0;
12         }
13     }
14     cout << "prime";
15     return 0;
16 }
17
```

7

OUTPUT

prime

3. GCD

GCD-Fun. ▾

GCD-FUN.

Write a function to determine the GCD (greatest common divisor) of two given integers.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,b;
5     cin>>a>>b;
6     while(b!=0){
7         int t = b;
8         b=a%b;
9         a=t;
10    }
11    cout<<a;
12    return 0;
13 }
```

OUTPUT

1

24

4. Reverse

Reverse ▾

REVERSE

Write a function to reverse a given string.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,rev = 0;
5     cin>>n;
6     while(n>0){
7         rev=rev*10+n%10;
8         n/=10;
9     }
10    cout<<rev;
11    return 0;
12 }
```

OUTPUT

54321

12345

5. Count

Count

COUNT
Write a function to count the number of words in a given string.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,count=0;
5     cin>>n;
6     while(n>0){
7         count++;
8         n/=10;
9     }
10    cout<<count;
11    return 0;
12 }
```

OUTPUT

5

12345

6. Min - Max

Min-Max

MIN-MAX
Write a function to find the minimum and maximum elements in a given array.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a,b;
5     cin>>a>>b;
6     if(a>b)
7         cout<<"max="<<a<<"min"<<b;
8     else
9         cout<<"max="<<b<<"min="<<a;
10    return 0;
11 }
12
```

OUTPUT

max=20min=10

10 20

7. Palindrome

Palindrome-Fun. ▾

PALINDROME-FUN.

Write a function to check if a given string is a palindrome or not.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n, temp, rev=0;
5     cin>>n;
6     temp=n;
7     while(n>0){
8         rev=rev*10+n%10;
9         n/=10;
10    }
11    if(temp==rev)
12        cout<<"palindrome";
13    else
14        cout<<"not a palindrome";
15    return 0;
16 }
```

OUTPUT

palindrome

1221

8. Area

Area ▾

AREA

Write a function to calculate the area of a circle given its radius.

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     float r;
5     cin>>r;
6     cout<<3.14*r*r;
7     return 0;
8 }
```

OUTPUT

78.5

5

9. Convert

Convert

CONVERT

Write a function to convert a given temperature from Celsius to Fahrenheit.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     float c;
5     cin >> c;
6     cout<<(c*9/5)+32;
7     return 0;
8 }
```

OUTPUT

93.2

34

10. Fibonacci

Fibonacci

FIBONACCI

Write a program to read in an integer and print the Fibonacci series up to that number using a for loop.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n,a=0,b=1,c;
5     cin>>n;
6     for(int i=1;i<=n;i++){
7         cout<<a<<" ";
8         c=a+b;
9         a=b;
10        b=c;
11    }
12    return 0;
13 }
```

OUTPUT

011235

6